

Nathan Xia

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ABOUT

First year Engineering Science student with experience in CFD, CAD, and aerodynamic design through national science competitions, design teams and research groups. Interested in aerodynamics, fluid control, and numerical simulations.

EDUCATION

- University of Toronto** Toronto, Canada
 - Bachelor of Applied Science - Engineering Science | Lachlan Dales McKeller Scholar* *September 2025 - Present*
 - cGPA: 3.95/4.00

SKILLS SUMMARY

- Languages:** Python, Java, MATLAB, C, Arduino API
- Tools:** ANSYS Fluent, Autodesk CFD, SolidWorks, Fusion 360, MATLAB, OpenRocket, OpenVSP/VSPAERO
- Skills:** CFD simulation and meshing, aerodynamic analysis and design, CAD, wind tunnel analysis and design

EXPERIENCE

- FCET Research Group** UTIAS
 - Research Intern* *May 2026 - September 2026*
 - Selected as one of only two first-year students to conduct research at UTIAS, independently leading an aeroacoustics study on remote microphone probe calibration for wind-tunnel pressure measurements.
 - Owned the full experimental workflow from apparatus and test-matrix design through data acquisition, FFT-based signal processing, statistical analysis, and technical presentation at UnERD across 16 investigated variables.
 - Developed a MATLAB acquisition and analysis pipeline and identified calibration and geometry improvements
 - Produced experimentally validated findings that informed the group's ongoing RMP research and future aeroacoustic measurement practices, including improved calibration configurations and procedures.
- Robotics For Space Exploration - Aerial** University of Toronto
 - Mechanical Co-Lead* *July 2026 - Present*
 - Directed all mechanical design and integration activities for the subsystem, with responsibility for architecture, CAD, analysis, manufacturing, assembly, and final system integration.
 - Led a team of X mechanical recruits, overseeing CAD onboarding, CFD analysis, design iteration, and manufacturing while reviewing and coordinating technical work across the team.
 - Mechanical Sub-team Recruit* *Sept 2025 - July 2026*
 - Conceived the glider's lifting-body configuration, replacing a conventional fuselage with an airfoil-profile body; evaluated 20+ airfoils in OpenVSP/VSPAERO to select and refine the final geometry.
 - Integrated passive flow-control devices, including vortex generators and end plates, to mitigate boundary-layer separation, increasing simulated lift by $5\times$ at 20 m/s.
 - Designed and simulated rocket nose-cone geometries in Autodesk CFD, iterating the aerodynamic profile using CFD and rocket-performance simulations.
- STEM Fellowship REO** University of Calgary
 - Student Lab Intern* *March 2025*
 - Selected from a competitive pool of ≈ 150 applicants for an introductory research internship, collaborating with graduate researchers and professors on carbon-cycle and seawater CO₂ removal.
 - Performed chemical sample analysis using ion chromatography (IC), optical emission spectroscopy (OES), and scanning electron microscopy (SEM), and presented findings to an audience of ≈ 60 .
- CWSF Finalist** Youth Science Canada
 - Vortex Generator Project* *December 2024 - Jun 2025*
 - 1 of 15 projects selected to compete in CWSF out of ≈ 600 total projects.
 - Bronze Medal recipient at CWSF out of ≈ 350 projects.

PROJECTS

- FCET Research Project:** Conducted a research project on the effects of remote microphone probe (RMP) geometry and calibration procedure on transfer functions (TF). Identified experimentally validated improvements that reduced transfer-function magnitude variability by 69%, phase variability by 33%, improved linearity by 46%, and increased sub-300 Hz coherence by 4.6%. All data acquisition was done through a custom MATLAB script. The large dataset collected provides a basis for improved RMP geometries in the field of aeroacoustics. [Link to the paper: found here.](#)
- Vortex Generator Research Project:** Conducted a research project optimizing vortex generators (VGs) for boundary layer separation mitigation on a NACA 2412 airfoil. Identified optimal VG long gap width and placement using a homemade flow visualization wind tunnel, to significantly increase stall angle and improve aerodynamic performance. Airfoil was mathematically designed on Desmos before importing into CAD software to be edited. Presented findings at CYSF and CWSF. [Link to the paper: found here.](#)
- Citation Machine:** Citation machine with functional UI written in Java. Able to generate citations in APA, MLA, and Chicago. Works for PostMedia, TorStar, and Canadian Encyclopedia branded websites. Involves web scraping through Java. [GitHub for the project: found here.](#)